```
#VENKATESH GAURI SHANKAR
```

```
#Exp 6: Implementation of data transformation using various methods.
```

```
import numpy as np
import pandas as pd
```

```
df = pd.read_csv("data.csv")
```

```
df
```

| | Cust id | Age | Annual Income (K$) | Spending Score (1-100) |
|---|---|---|---|---|
| **0** | 1 | 19 | 15 | 39 |
| **1** | 2 | 21 | 15 | 81 |
| **2** | 3 | 20 | 14 | 6 |
| **3** | 4 | 23 | 16 | 77 |
| **4** | 5 | 18 | 17 | 40 |

```
#Add a column
```

```
df['new'] = np.random.random(5)
```

```
df
```

| | Cust id | Age | Annual Income (K$) | Spending Score (1-100) |
|---|---|---|---|---|
| **0** | 1 | 19 | 15 | 39 |
| **1** | 2 | 21 | 15 | 81 |
| **2** | 3 | 20 | 14 | 6 |
| **3** | 4 | 23 | 16 | 77 |
| **4** | 5 | 18 | 17 | 40 |

```
df.drop('new', axis=1, inplace=True)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
<ipython-input-13-4065f14dd5a3> in <module>()
----> 1 df.drop('new', axis=1, inplace=True)
```

```
df
```

| | Cust id | Age | Annual Income (K$) | Spending Score (1-100) |
|---|---|---|---|---|
| **0** | 1 | 19 | 15 | 39 |
| **1** | 2 | 21 | 15 | 81 |
| **2** | 3 | 20 | 14 | 6 |
| **3** | 4 | 23 | 16 | 77 |
| **4** | 5 | 18 | 17 | 40 |

```
#Add a row
```

```
df.loc[5,:] = [6, 20, 13, 45]
```

```
df
```

| | Cust id | Age | Annual Income (K$) | Spending Score (1-100) |
|---|---|---|---|---|
| **0** | 1.0 | 19.0 | 15.0 | 39.0 |
| **1** | 2.0 | 21.0 | 15.0 | 81.0 |
| **2** | 3.0 | 20.0 | 14.0 | 6.0 |
| **3** | 4.0 | 23.0 | 16.0 | 77.0 |
| **4** | 5.0 | 18.0 | 17.0 | 40.0 |
| **5** | 6.0 | 20.0 | 13.0 | 45.0 |

```
df.drop(5, axis=0, inplace=True)
```

```
df
```

| | Cust id | Age | Annual Income (K$) | Spending Score (1-100) |
|---|---|---|---|---|
| **0** | 1.0 | 19.0 | 15.0 | 39.0 |
| **1** | 2.0 | 21.0 | 15.0 | 81.0 |
| **2** | 3.0 | 20.0 | 14.0 | 6.0 |
| **3** | 4.0 | 23.0 | 16.0 | 77.0 |
| **4** | 5.0 | 18.0 | 17.0 | 40.0 |

```
#Using Insert function
```

```
df.insert(0, 'new', np.random.random(5))
```

```
df
```

| | new | Cust id | Age | Annual Income (K$) | Spending Score (1-100) |
|---|---|---|---|---|---|
| **0** | 0.766037 | 1.0 | 19.0 | 15.0 | 39.0 |
| **1** | 0.304991 | 2.0 | 21.0 | 15.0 | 81.0 |
| **2** | 0.099474 | 3.0 | 20.0 | 14.0 | 6.0 |
| **3** | 0.352563 | 4.0 | 23.0 | 16.0 | 77.0 |
| **4** | 0.280371 | 5.0 | 18.0 | 17.0 | 40.0 |

```
df.drop('new', axis=1, inplace=True)
```

```
df
```

| | Cust id | Age | Annual Income (K$) | Spending Score (1-100) |
|---|---|---|---|---|
| **0** | 1.0 | 19.0 | 15.0 | 39.0 |
| **1** | 2.0 | 21.0 | 15.0 | 81.0 |
| **2** | 3.0 | 20.0 | 14.0 | 6.0 |
| **3** | 4.0 | 23.0 | 16.0 | 77.0 |
| **4** | 5.0 | 18.0 | 17.0 | 40.0 |

```
#Melt function
```

```
import pandas as pd
df1 = pd.read_csv("data1.csv")
```

```
df1
```

| | names | day1 | day2 | day3 | day4 | day5 |
|---|---|---|---|---|---|---|
| **0** | jane | 1 | 5 | 8 | 5 | 3 |
| **1** | john | 5 | 1 | 3 | 2 | 8 |
| **2** | jack | 4 | 9 | 8 | 6 | 3 |

```
pd.melt(df1, id_vars='names').head()
```

| | names | variable | value |
|---|---|---|---|
| 0 | jane | day1 | 1 |

```
#Concat function
```

```
import pandas as pd
df2 = pd.read_csv("data2.csv")
```

| 4 | john | day2 | 1 |
|---|---|---|---|

```
df2
```

| | names | day1 | day2 | day3 | day4 | day5 |
|---|---|---|---|---|---|---|
| 0 | max | 1 | 4 | 6 | 4 | 2 |
| 1 | mini | 6 | 0 | 2 | 1 | 5 |
| 2 | mily | 3 | 3 | 7 | 7 | 0 |

```
pd.concat([df2, df1], axis=0, ignore_index=True)
```

| | names | day1 | day2 | day3 | day4 | day5 |
|---|---|---|---|---|---|---|
| 0 | max | 1 | 4 | 6 | 4 | 2 |
| 1 | mini | 6 | 0 | 2 | 1 | 5 |
| 2 | mily | 3 | 3 | 7 | 7 | 0 |
| 3 | jane | 1 | 5 | 8 | 5 | 3 |
| 4 | john | 5 | 1 | 3 | 2 | 8 |
| 5 | jack | 4 | 9 | 8 | 6 | 3 |

```
#merge function
```

```
import pandas as pd
df3 = pd.read_csv("data3.csv")
```

```
df3
```

| | names | rating |
|---|---|---|
| 0 | monk | 1 |
| 1 | mini | 2 |
| 2 | mily | 3 |

```
df2.merge(df3, on='names')
```

```
df3.merge(df2, on='names')
```

| | names | rating | day1 | day2 | day3 | day4 | day5 |
|---|---|---|---|---|---|---|---|
| **0** | mini | 2 | 6 | 0 | 2 | 1 | 5 |
| **1** | mily | 3 | 3 | 3 | 7 | 7 | 0 |

```
# Get dummies to handle categorical variables
```

```
import pandas as pd
df4 = pd.read_csv("data4.csv")
```

```
df4
```

| | name | ctg | vals |
|---|---|---|---|
| **0** | monk | A | 10 |
| **1** | mini | A | 15 |
| **2** | mily | C | 20 |
| **3** | James | B | 14 |
| **4** | John | B | 18 |

```
pd.get_dummies(df4)
```

| | vals | name_James | name_John | name_mily | name_mini | name_monk | ctg_A | ctg_B | ctg_C |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 10 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| **1** | 15 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| **2** | 20 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| **3** | 14 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| **4** | 18 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |

```
# To transform Pivot table for aggregation
```

```
import pandas as pd
df5 = pd.read_csv("data5.csv")
```

```
df5
```

| | name | ctg | vals |
|---|---|---|---|
| **0** | monk | A | 10 |
| **1** | mini | A | 15 |
| **2** | mily | C | 20 |
| **3** | mily | A | 13 |
| **4** | mini | C | 17 |

```
df5.pivot_table(index='name', columns='ctg', aggfunc='mean')
```

| | vals | | |
|---|---|---|---|
| **ctg** | **A** | **B** | **C** |
| **name** | | | |
| **mily** | 13 | 10 | 20 |
| **mini** | 15 | 20 | 17 |
| **monk** | 10 | 11 | 16 |

```
# .query() for serach transformation
```

```
df5.query('vals== 20 | vals == 10')
```

| | name | ctg | vals |
|---|---|---|---|
| **0** | monk | A | 10 |
| **2** | mily | C | 20 |
| **6** | mily | B | 10 |
| **7** | mini | B | 20 |

```
# .sort_values() for sorting transformation
```

```
df5.sort_values(by = ['vals'])
```

| | name | ctg | vals |
|---|---|---|---|

```
# .filter() transformation
```

```
df5.filter(['ctg'])
```

| | ctg |
|---|---|
| **0** | A |
| **1** | A |
| **2** | C |
| **3** | A |
| **4** | C |
| **5** | B |
| **6** | B |
| **7** | B |
| **8** | C |